6F BOND ENTHALPY AND MEAN BOND ENTHALPY

SPECIFICATION REFERENCE 6.9 PART

LEARNING OBJECTIVES

- Understand the terms bond enthalpy and mean bond enthalpy

WHAT IS BOND ENTHALPY?

Bond enthalpy ($\Delta_B H$) is the enthalpy change when one mole of a bond in the gaseous state is broken.

For a *diatomic* molecule, XY, the bond enthalpy is the enthalpy change for the following reaction:

$$XY(g) \rightarrow X(g) + Y(g)$$

Some examples are:

$$Cl_2(g) \rightarrow 2Cl(g) \qquad \Delta_B H = +243\,kJ\,mol^{-1}$$

$$H_2(g) \rightarrow 2H(g) \qquad \Delta_B H = +436\,kJ\,mol^{-1}$$

$$HCl(g) \rightarrow H(g) + Cl(g) \qquad \Delta_B H = +432\,kJ\,mol^{-1}$$

For *polyatomic* molecules, each bond has to be considered separately. For example, with methane there are four separate bond enthalpies:

$$CH_4(g) \rightarrow CH_3(g) + H(g) \qquad \Delta_B H = +423\,kJ\,mol^{-1}$$

$$CH_3(g) \rightarrow CH_2(g) + H(g) \qquad \Delta_B H = +480\,kJ\,mol^{-1}$$

$$CH_2(g) \rightarrow CH(g) + H(g) \qquad \Delta_B H = +425\,kJ\,mol^{-1}$$

$$CH(g) \rightarrow C(g) + H(g) \qquad \Delta_B H = +335\,kJ\,mol^{-1}$$

WHAT IS MEAN BOND ENTHALPY?

You will notice that the bond enthalpy of the C—H bond varies with its environment. For this reason it is often useful to quote the **mean bond enthalpy**.

The mean bond enthalpy for the C—H bond in methane is approximately $+416\,kJ\,mol^{-1}$.

$$\left[\tfrac{1}{4}(423 + 480 + 425 + 335) = 415.75\right]$$

The mean bond enthalpy for the C—H bond in a large number of organic compounds is $+413\,kJ\,mol^{-1}$. If the bond enthalpy is calculated for a particular compound, it will probably be slightly different from the mean value. For example, the mean bond enthalpy for the C—H bond in ethane is $+420\,kJ\,mol^{-1}$.

Some examples of mean bond enthalpies are given in **table A**.

BOND	MEAN BOND ENTHALPY / kJ mol^{-1}	BOND	MEAN BOND ENTHALPY / kJ mol^{-1}
C–C	+347	O–H	+464
C=C	+612	C–F	+467
C≡C	+838	C–Cl	+346
C–O	+358	C–Br	+290
C=O	+743	C–I	+228

table A Examples of mean bond enthalpies.

A quick representation for mean bond enthalpy is to use the letter *E* followed by the bond in brackets. So, the mean bond enthalpy of the C—C bond is written as E(C—C) = $+347\,kJ\,mol^{-1}$.

CHECKPOINT

SKILLS PROBLEM SOLVING

1. (a) State what we mean by the H–I bond enthalpy.

(b) Write an equation that represents the bond enthalpy of the H–I bond.

2. The two bonds in the water molecule can be separately broken. The enthalpy changes for the two processes are:

$$H–O–H(g) \rightarrow O–H(g) + H(g) \qquad \Delta H = +496\,kJ\,mol^{-1}$$

$$O–H(g) \rightarrow O(g) + H(g) \qquad \Delta H = +432\,kJ\,mol^{-1}$$

(a) Calculate the mean bond enthalpy for the O–H bond in water.

(b) Why is this value not the same as the value for the mean bond enthalpy of an O–H bond, which is quoted in a data book?

3. Suggest a reason why the mean bond enthalpy of the C–H bond in methane ($+416\,kJ\,mol^{-1}$) is different from the one in ethane ($+420\,kJ\,mol^{-1}$).

4. (a) Write equations to show the successive breaking of the N–H bonds in ammonia, $NH_3(g)$.

(b) Explain how the enthalpy changes for each reaction are used to calculate the mean bond enthalpy for the N–H bond in ammonia.

SUBJECT VOCABULARY

bond enthalpy the enthalpy change when one mole of a bond in the gaseous state is broken

mean bond enthalpy the enthalpy change when one mole of a bond, averaged out over many different molecules, is broken